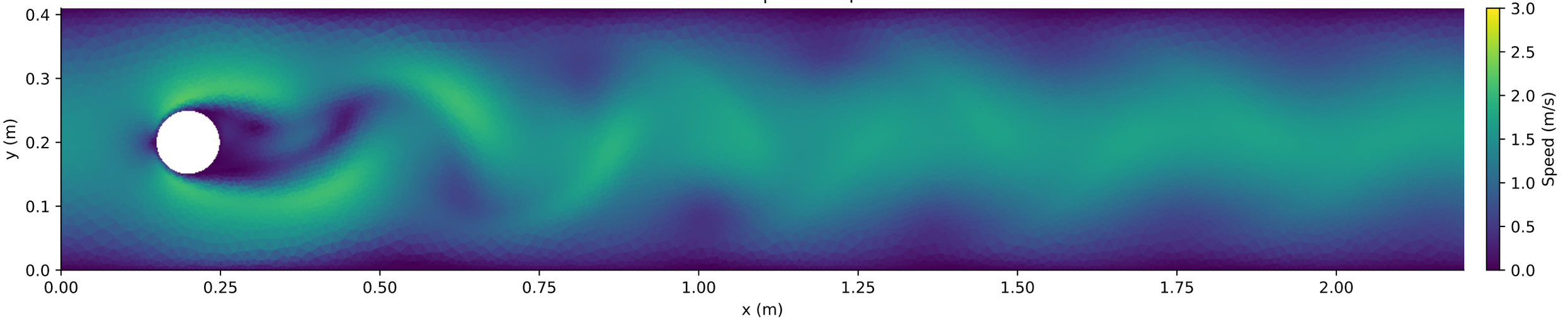
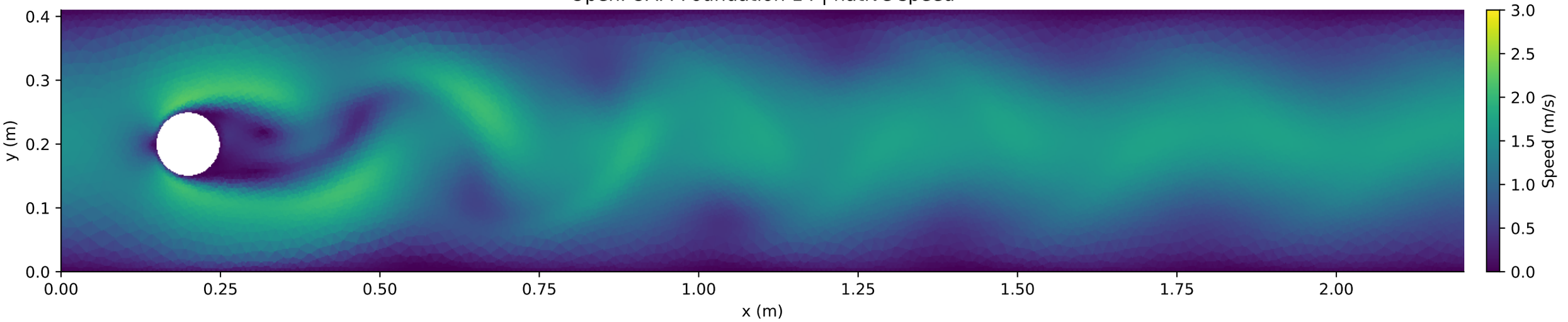
Cylinder wake | physical time t = 4.000 s

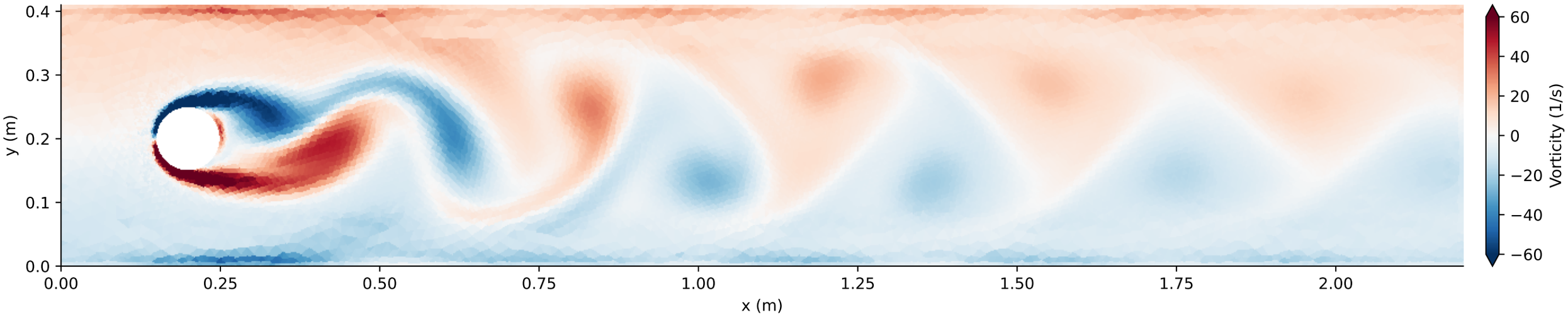
ANSYS Fluent | native speed



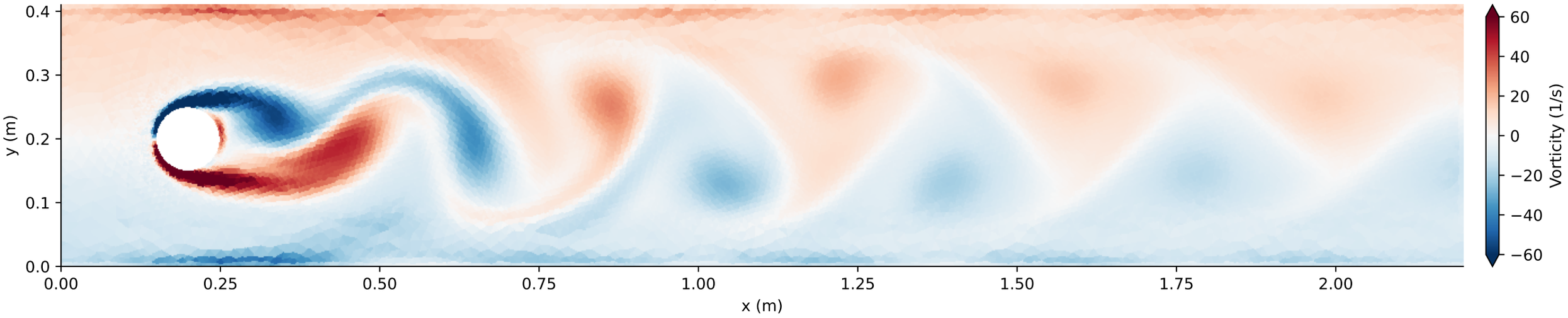
OpenFOAM Foundation 14 | native speed



ANSYS Fluent | signed vorticity from native velocity



OpenFOAM Foundation 14 | signed vorticity from native velocity



21,598 cells | Re = 100 | dt = 0.005 s | fixed scales across both codes and all frames
800 new steps (4–8 s) meet their declared numerical checks; inherited startup remains. No phase alignment.
Vorticity is an area-weighted velocity-gradient reconstruction, clipped at +/-60 1/s; native values remain unchanged.
Actual native states only. No generated intermediate motion. Benchmark matching and physical validation remain open.